



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY



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2024

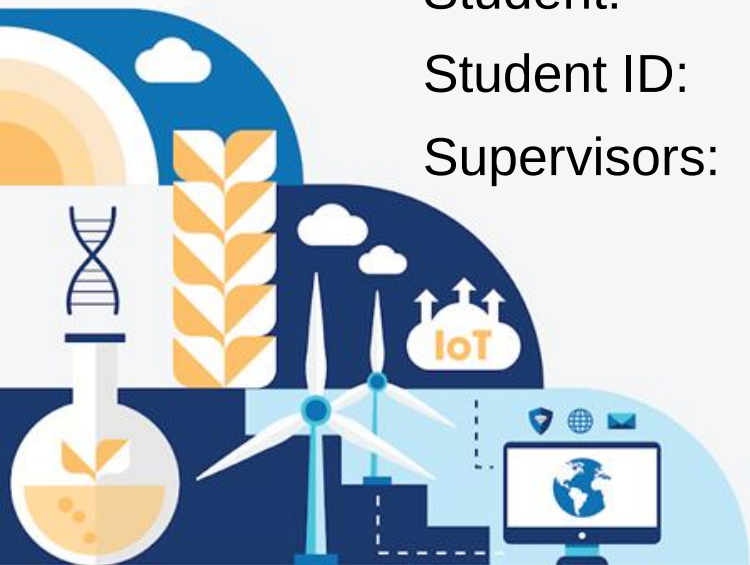
MỘT BÁCH KHOA
KẾT NỐI - SÁNG TẠO - XUẤT SẮC



Student Research and Creativity Conference for the
2023-2024 School Year

Topic: Attack and defense in 6G-IoT network exploiting artificial intelligence

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The novelty of the research

The novelties of our research are summarized as follows:

- We propose 3 algorithms for generating adversarial perturbation in a finite scattering environment close to the real environment.
- We consider the MIMO-RIS-based autoencoder in a finite scattering environment under adversarial attack.
- We highlight the adversarial attack and defence approaches to various technologies in 6G-IoT systems.



Outline and presentation schedule

I, Introduction (2 mins)

II, Related work (2 mins)

III, Problem formulation (2 mins)

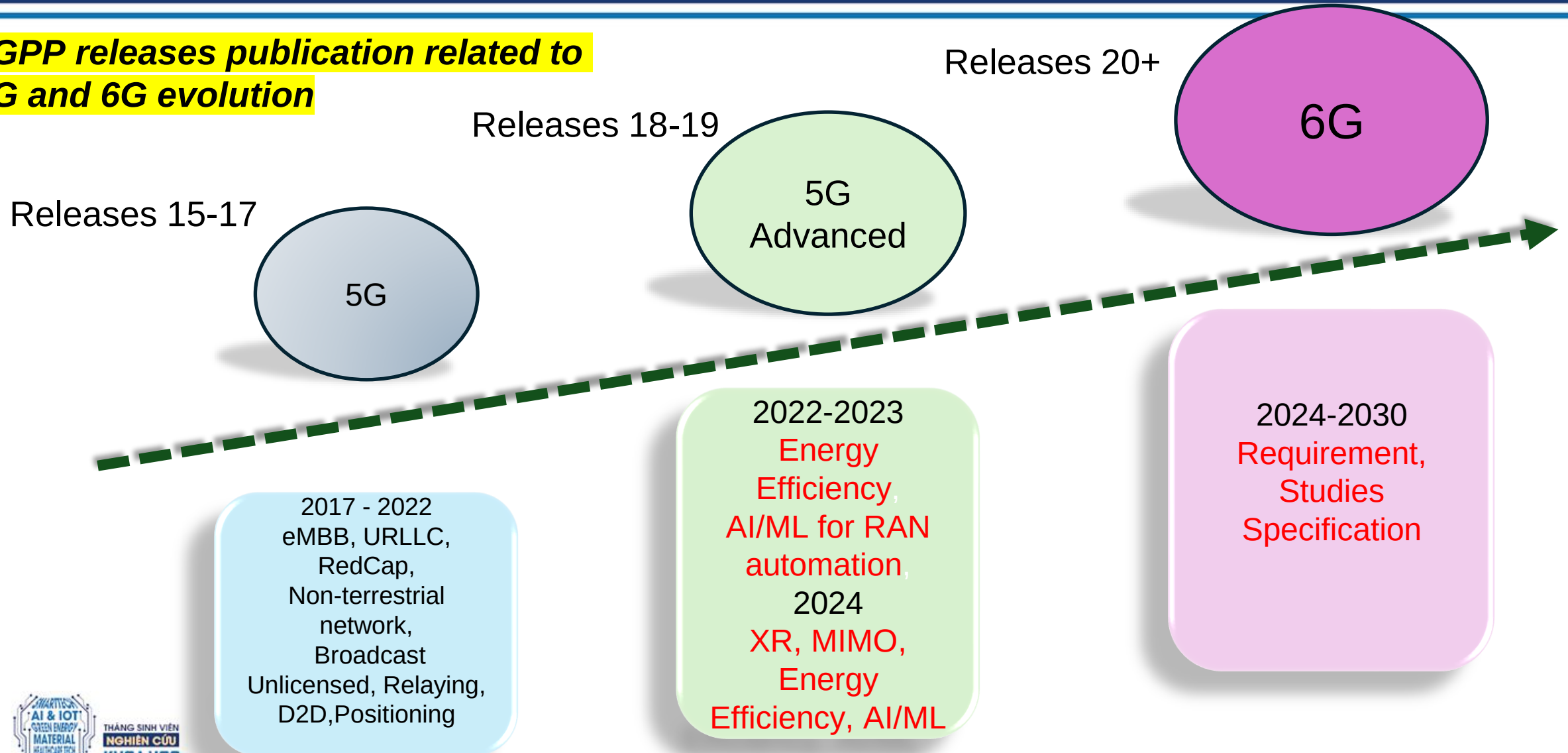
IV, System model (4 mins)

V, Numerical results (5 mins)

VI, Conclusion(1 min)

I. Introduction

3GPP releases publication related to 5G and 6G evolution



I. Introduction

Communication theory + AI theory = Theory of communication system-based AI

Hardware improvement + User requirement = IoT Devices

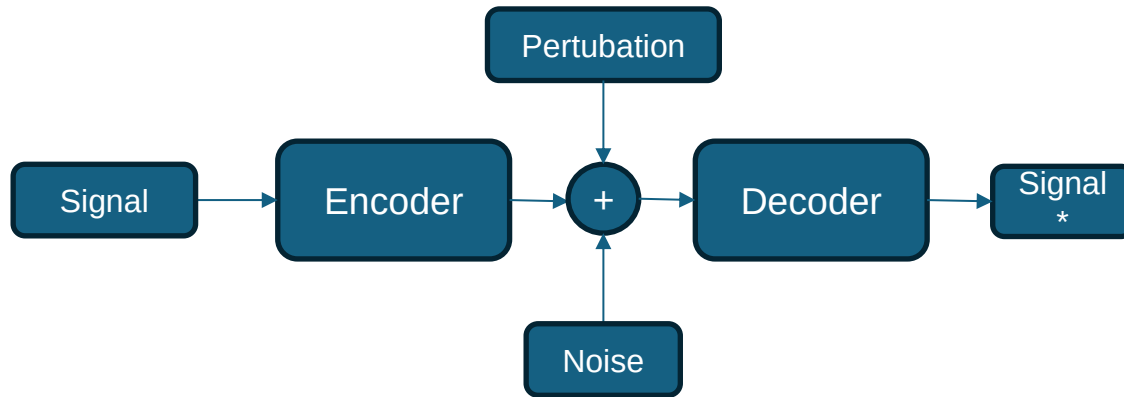
Protocols + IoT devices = IoT networks



Theory of communication system-based AI + IoT networks + Wireless communication system = 6G-IoT Network-based AI

II. Related work

Research problems: The performance of adversarial attack compared to jamming attack in simple communication systems.

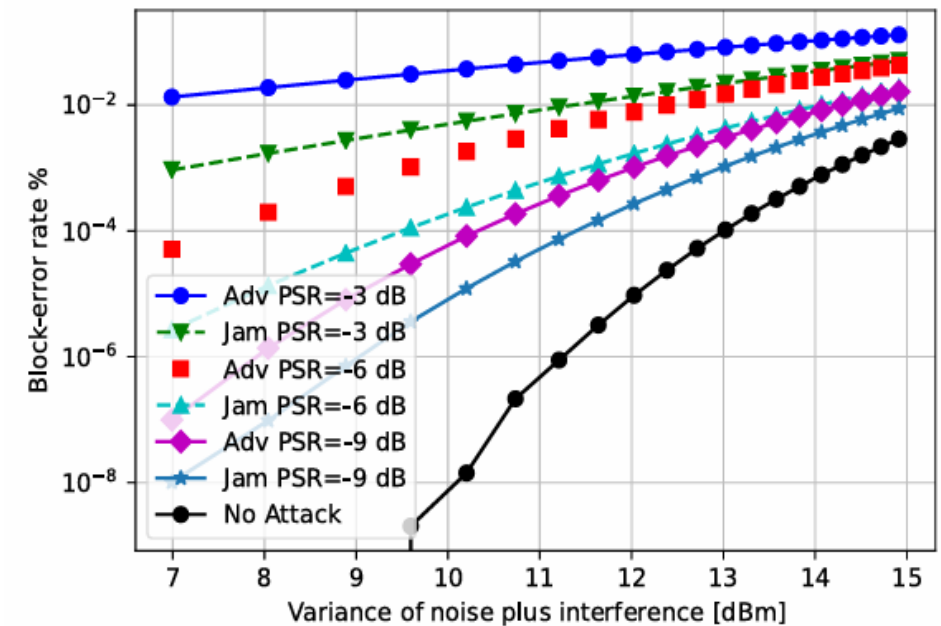


Simple autoencoder in AWGN Channel [3]

[3] M. Sadeghi and E. G. Larsson, "Physical Adversarial Attacks Against End-to-End Autoencoder Communication Systems," in *IEEE Communications Letters*, vol. 23, no. 5, pp. 847-850, May 2019, doi: 10.1109/LCOMM.2019.2901469

Major research contributions:

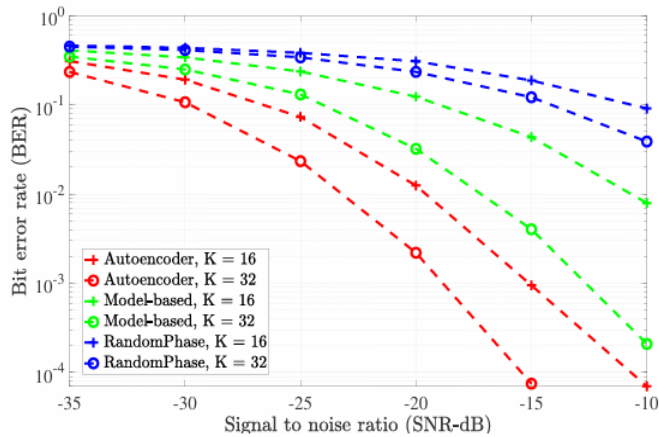
- ✓ A **basic adversarial attack algorithm** is **more effective** than a **jamming attack** employing AWGN.
- ✓ An **simple autoencoder** can be **damaged** by both jamming and adversarial attack.



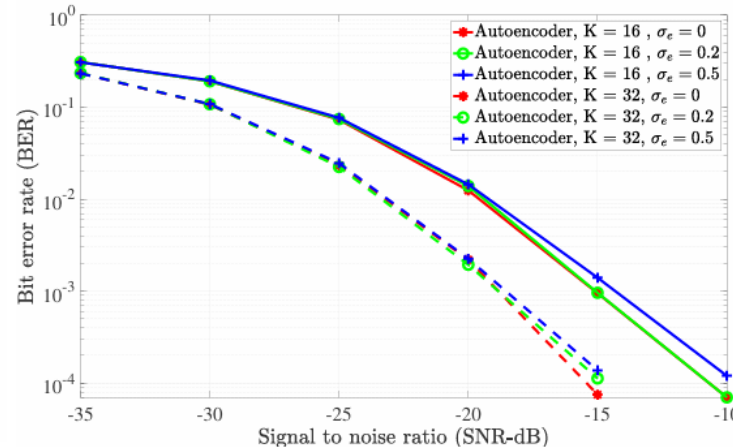
Simulation result with different configuration [1]

II. Related work

Authors in [4] proposed a RIS-MIMO-based autoencoder



The BER performance [4]



Imperfect channel state information [4]

[4] H. A. Le, T. Van Chien, V. D. Nguyen and W. Choi, "RIS-Assisted MIMO Communication Systems: Model-based versus Autoencoder Approaches," 2022 IEEE 33rd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Kyoto, Japan, 2022, pp. 1-6, doi: 10.1109/PIMRC54779.2022.9977540.

Research problems: The performance of RIS-MIMO-based autoencoder.

Major research contributions:

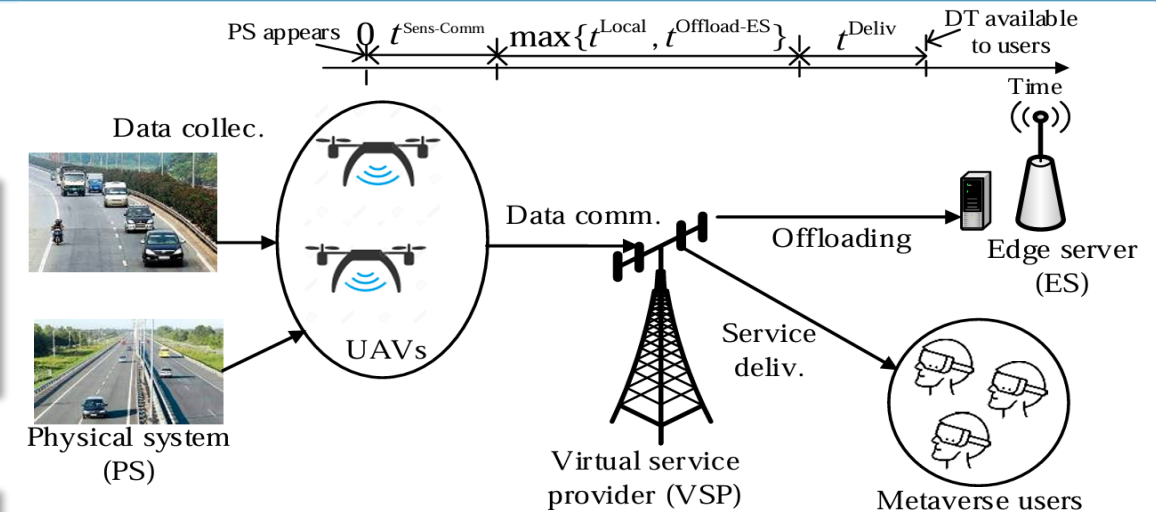
- ✓ The proposed model was suitable to encode the transmitted signal.
- ✓ The proposed model gained BER performance compared to other state-of-the-art models
- ✓ The proposed model ensured system performance under imperfect channel information cases

III. Problem formulation

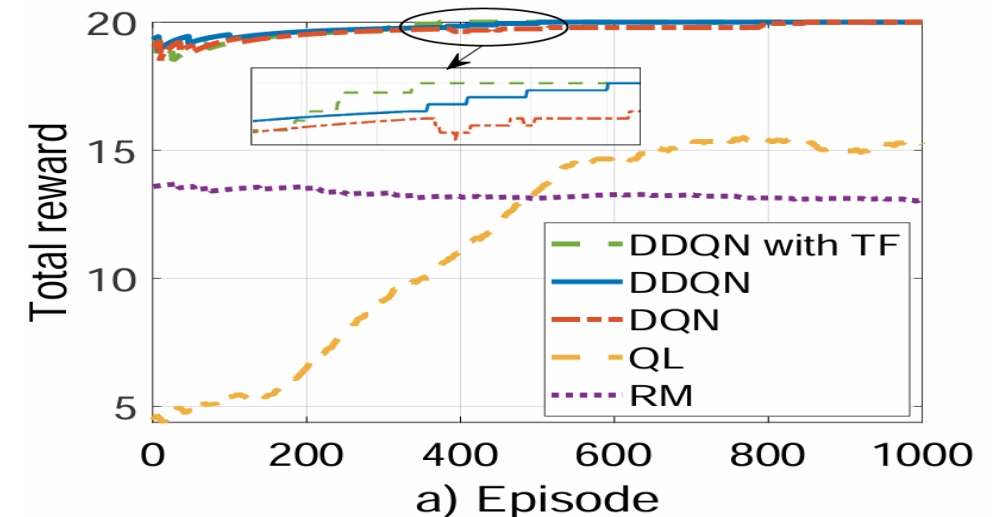
Research problems: Deep learning is suitable for processing dynamic data in key technologies and applications in 6G [2].

Major contributions

- ✓ The trained DRL algorithms can be used in real-time to satisfy the latency requirements of Metaverse users.
- ✓ We consider the dynamics and uncertainty of both the wireless channel state between the UAVs and the VSP and the channel condition between the VSP and the ES

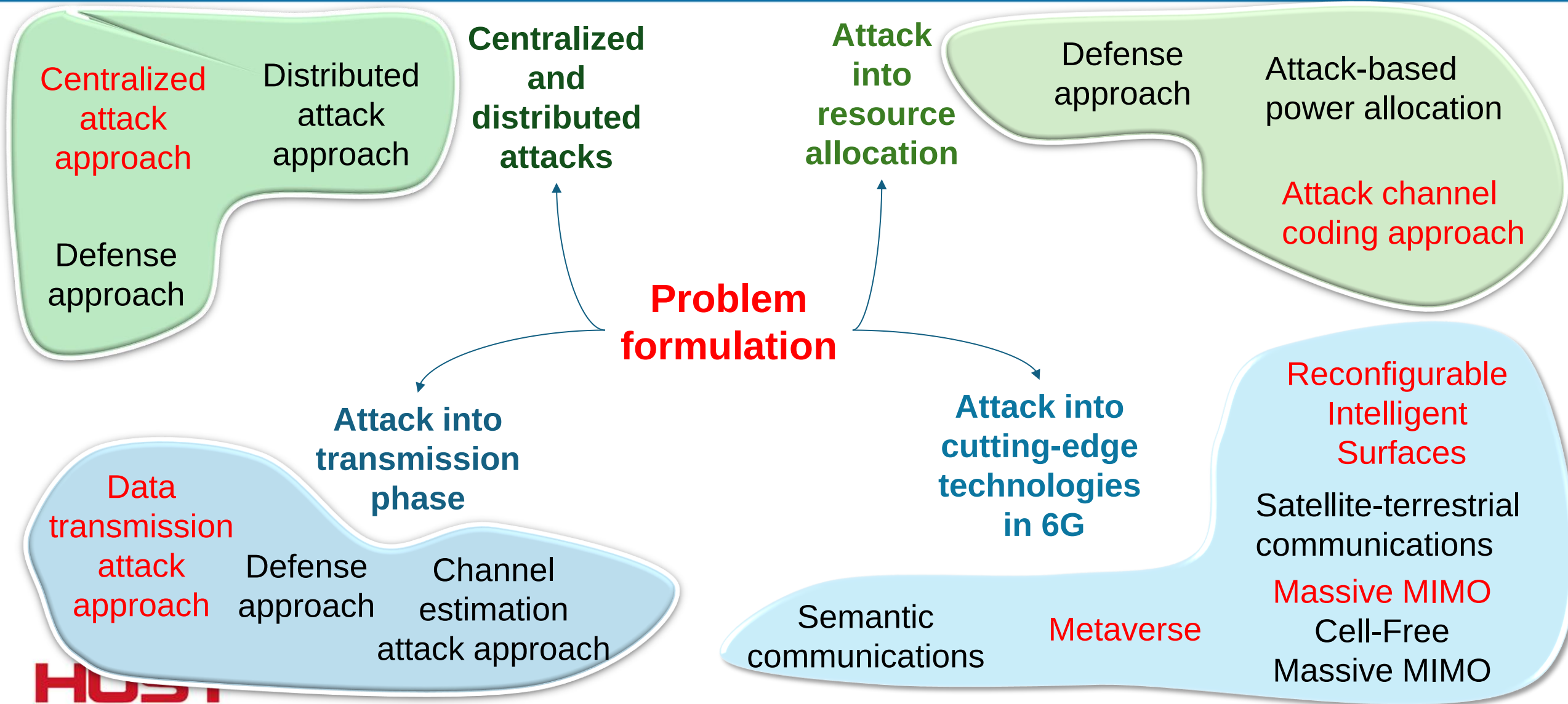


An edge computing-assisted Metaverse system [2]



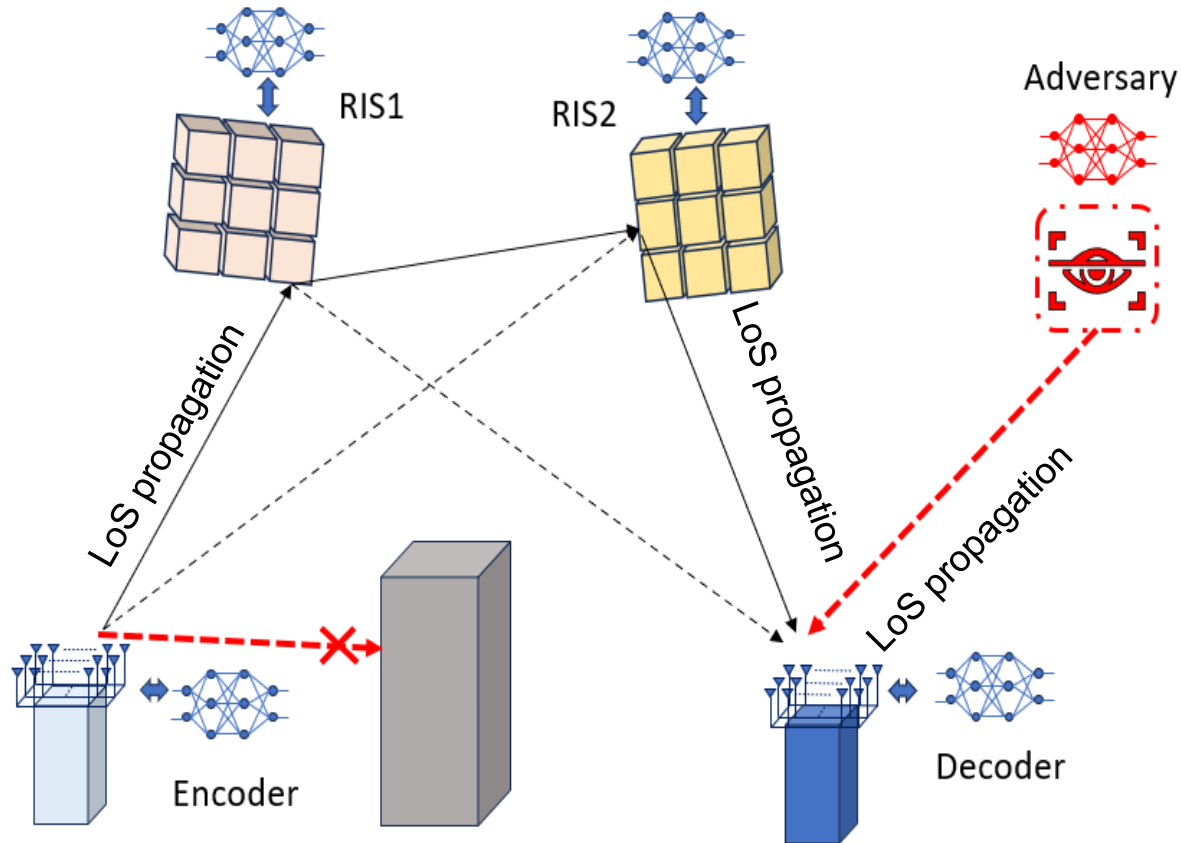
Convergence of the algorithms [2]

III, Problem formulation



IV. System model

4.1 Model architecture



System model

Research problems: Constructing adversarial attack against a double RIS-assisted MIMO-based Autoencoder in finite scattering environment .

Major contributions:

- ✓ We propose a system model including five components to **create end-to-end learning system** in finite scattering environment.
- ✓ We **successfully construct an adversary** in white box scenario.
- ✓ We exploit **1D-CNN** to **replace physical component** of communication system.

B. D. Son et al., "Adversarial Attacks and Defenses in 6G Network-Assisted IoT Systems" in *IEEE Internet of Things Journal* **Q1-IF =10.6**

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IV. System model

4.2 Proposed algorithms

Algorithm 1 Producing RIS-MIMO adversarial example-based FGM

Input: w , Model architecture, ϵ , $\mathbf{p}_{\text{PSR}}$, y_{label}

```
1: INIT:  $\epsilon = \mathbf{0}^{M_Q \times B_L}$ 
2: for  $i$  in range( $M_Q \times B_L$ ) do
3:    $\epsilon_{\min} = 0$  and  $\epsilon_{\max} = p_{\max}$ 
4:    $\mathbf{p}_{\text{norm}} = \frac{\nabla_w \mathcal{L}(w, e_i)}{(\|\nabla_w \mathcal{L}(w, e_i)\|_2)}$ 
5:   while  $\epsilon_{\max} - \epsilon_{\min} > \epsilon_{\text{acc}}$  do
6:      $\epsilon_{\text{ave}} = (\epsilon_{\max} + \epsilon_{\min})/2$ 
7:      $w_{\text{adv}} = w - \epsilon_{\text{ave}} \mathbf{p}_{\text{norm}}$ 
8:     if  $y_{\text{pred}} == y_{\text{label}}$  then
9:        $\epsilon_{\min} = \epsilon_{\text{ave}}$ 
10:    else
11:       $\epsilon_{\max} = \epsilon_{\text{ave}}$ 
12:    end if
13:  end while
14:   $\epsilon_i = \epsilon_{\max}$ 
15: end for
16:  $\text{target} = \arg \min \epsilon$  and  $\epsilon^* = \min \epsilon$ 
17:  $\mathbf{p}_{\text{add}} = \frac{\epsilon^* \nabla_w \mathcal{L}(w, e_{\text{target}})}{(\|\nabla_w \mathcal{L}(w, e_{\text{target}})\|_2)}$ 
Output:  $\mathbf{p}_{\text{add}}$ 
```

Algorithm 2 Producing RIS-MIMO adversarial example-based PGD

Input: w , Model architecture, ϵ , $\mathbf{p}_{\text{PSR}}$, y_{label}

```
1: INIT:  $\epsilon = \mathbf{0}^{M_Q \times B_L}$ 
2: for  $i$  in range( $M_Q \times B_L$ ) do
3:    $\epsilon_{\min} = 0$  and  $\epsilon_{\max} = p_{\max}$ 
4:    $\mathbf{p}_{\text{norm}} = \frac{\nabla_w \mathcal{L}(w, e_i)}{(\|\nabla_w \mathcal{L}(w, e_i)\|_2)}$ 
5:   while  $\epsilon_{\max} - \epsilon_{\min} > \epsilon_{\text{acc}}$  do
6:      $\mathbf{p}_{\text{temp}} = \mathbf{p}_{\text{norm}}$  and  $w_{\text{adv}} = w$ 
7:     for  $j$  in range  $n_s$  do
8:        $\epsilon_{\text{ave}} = (\epsilon_{\max} + \epsilon_{\min})/2$ 
9:        $w_{\text{adv}} = w_{\text{adv}} - (\epsilon_{\text{ave}}/n_s) \mathbf{p}_{\text{temp}}$ 
10:       $w_{\text{adv}} = \Pi_{\beta}(w_{\text{adv}})$ 
11:       $\mathbf{p}_{\text{temp}} = \nabla_{w_{\text{adv}}} \mathcal{L}(w_{\text{adv}}, e_i) / \|\nabla_{w_{\text{adv}}} \mathcal{L}(w_{\text{adv}}, e_i)\|_2$ 
12:    end for
13:     $\mathbf{p}_{\text{norm}} = \mathbf{p}_{\text{temp}}$ 
14:    if  $D(w_{\text{adv}}) == y_{\text{label}}$  then
15:       $\epsilon_{\max} = \epsilon_{\text{ave}}$ 
16:    else
17:       $\epsilon_{\max} = \epsilon_{\text{ave}}$ 
18:    end if
19:  end while
20:   $\epsilon_i = \epsilon_{\max}$ 
21: end for
22:  $\text{target} = \arg \min \epsilon$  and  $\epsilon^* = \min \epsilon$ 
23:  $\mathbf{p}_{\text{add}} = \frac{\epsilon^* \nabla_w \mathcal{L}(w, e_{\text{target}})}{(\|\nabla_w \mathcal{L}(w, e_{\text{target}})\|_2)}$ 
Output:  $\mathbf{p}_{\text{add}}$ 
```

Algorithm 3 Formulating RIS-MIMO adversarial perturbations

Input: Model architecture, $\mathbf{p}_{\text{PSR}}$, σ^2

```
1: Set  $n_p = 50$  and  $\mathbf{p}_{\text{adv}} = \mathbf{0}$ 
2: for  $i$  in range  $n_p$  do
3:   Choose  $\mathbf{I}_r \in \mathbf{I}_D^{M_Q \times B_L}$  uniformly at random
4:   Set a random noise  $n \sim \mathcal{CN}(0, \sigma^2 \mathbf{I})$ 
5:   Set  $\hat{\mathbf{I}}_r = D(R_2(R_1(E(\mathbf{I}_r))) + n + \mathbf{p}_{\text{adv}}) = D(w_{\text{adv}})$ 
6:   if  $\hat{\mathbf{I}}_r = \mathbf{I}_r$  then
7:      $w_{\text{adv}} = w_{\text{adv}} + \mathbf{p}_{\text{adv}}$ 
8:     Apply algorithm 1 or 2 to calculate  $\mathbf{p}_{\text{add}}$ ,
9:     if  $\|\mathbf{p}_{\text{adv}} + \mathbf{p}_{\text{update}}\|_2^2 \leq \mathbf{p}_{\text{PSR}}$  then
10:       $\mathbf{p}_{\text{adv}} = \mathbf{p}_{\text{adv}} + \mathbf{p}_{\text{add}}$ 
11:    else
12:       $\mathbf{p}_{\text{adv}} = \sqrt{\mathbf{p}_{\text{PSR}}} \frac{\mathbf{p}_{\text{adv}} + \mathbf{p}_{\text{add}}}{\|\mathbf{p}_{\text{adv}} + \mathbf{p}_{\text{add}}\|_2}$ 
13:    end if
14:  end if
15: end for
Output:  $\mathbf{p}_{\text{adv}}$ 
```

V. Numerical results

Simulation parameters

Parameters	Value	Parameters	Value
Sample data	100000	Optimizer	Adam
SC	5,7,9	Learning rate	0.001
B_L	20	Loss function	Binary Cross Entropy Loss
Batch size	128	p_{PSR}	-7 dB
SNR train	15 dB	n_p	50
Number of epochs	1000	n_s	20
A_1	32	d_1	100m
A_2	32	d_2	200m
Modulation (M_Q)	64-QAM	d_H	2m
Channel	Rician Channel	N_{re}	16
Rician factor	0.2	N_{tr}	16

Configuration

Layer	Parameters	Output Dim
Encoder		
Input	None	$M \times L$
Conv1D + BN + ReLU	kernel = 1, filter = 256	$256 \times L$
Conv1D + BN + ReLU	kernel = 1, filter = 256	$256 \times L$
Conv1D + BN	kernel = 1, filter = 2 Nt	$256 \times L$
Decoder		
Input	None	$(2 N_r + 2 N_t N_r) M \times L$
Conv1D + BN + ReLU	kernel = 1, filter = 512	$512 \times L$
Conv1D + BN + ReLU	kernel = 1, filter = 512	$512 \times L$
Conv1D + Softmax	kernel = 1, filter = 2Nt	$M \times L$
RIS1 and RIS2		
Input	None	$2 K_i \times L$
Conv1D + BN + ReLU	kernel = 1, filter = 512	$512 \times L$
Conv1D + BN + ReLU	kernel = 1, filter = 512	$512 \times L$
Conv1D	kernel = 1, filter = K_i	$K_i \times L$

V. Numerical results

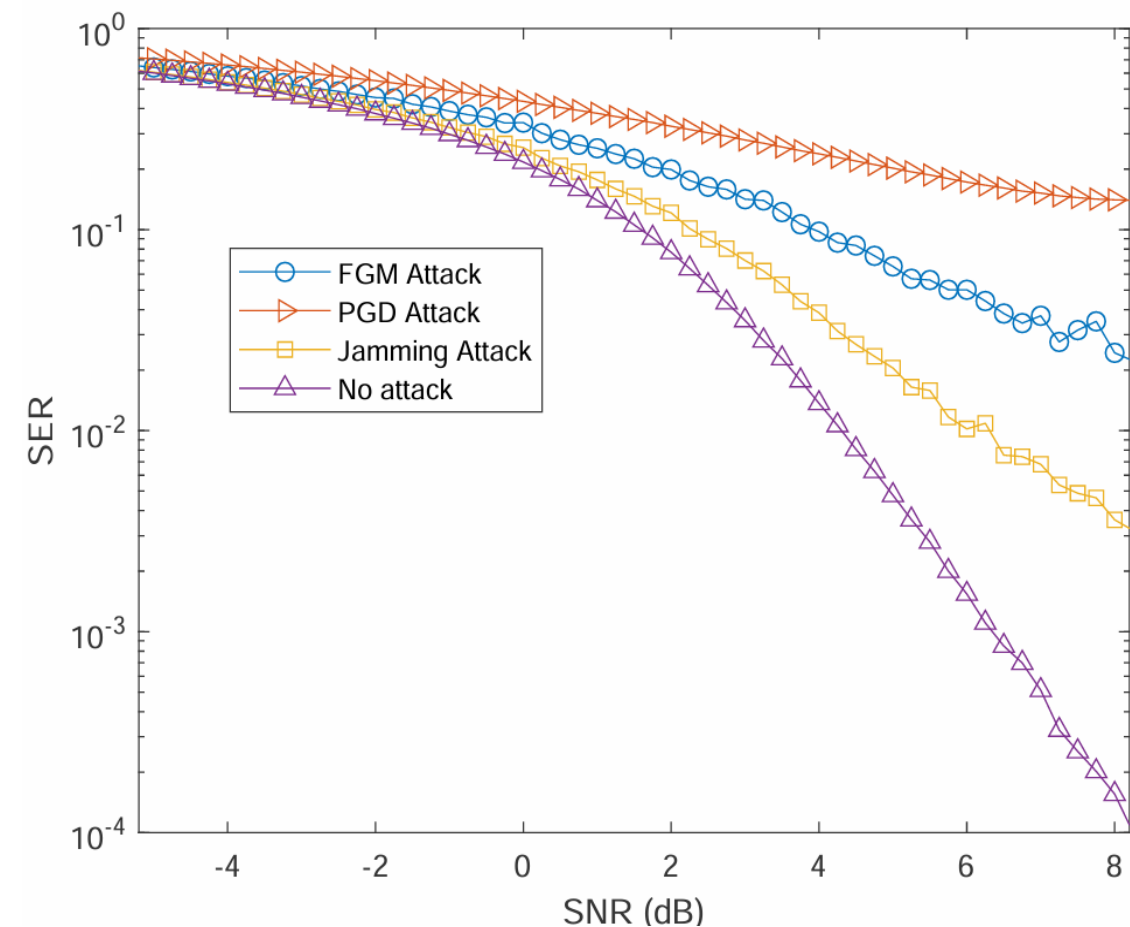
Research problems: Deep learning is suitable for processing dynamic data in key technologies and applications in 6G [2].

Major contributions:

- ✓ **PGD** is more **effective** at **deconstructing** than **FGM**
- ✓ Attack methods show their **ability** to **decrease** the **performance** of **RIS-MIMO-based AE**.
- ✓ **Without attack**, **AE** show its **effectiveness** in **minimize system's SER**

B. D. Son et al., "Adversarial Attacks and Defenses in 6G Network-Assisted IoT Systems" in *IEEE Internet of Things Journal* **Q1-IF =10.6**

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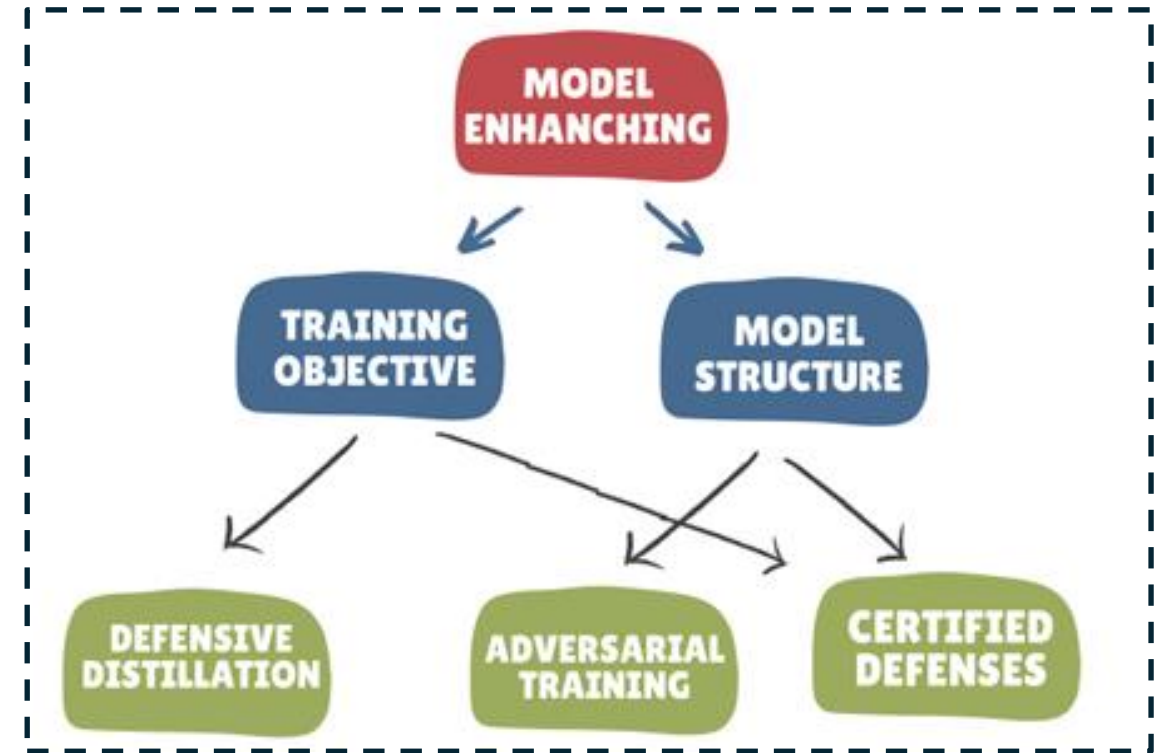
Simulation results

V. Numerical results

Research problems: Deep learning is suitable for processing dynamic data in key technologies and applications in 6G [2].

Protecting approaches:

- ✓ Defensive distillation aims to develop a more refined classifier that is better equipped to protect against adversarial attacks that rely on accurate gradient information.
- ✓ Certified defense involves the integration of a noise layer into the baseline model, generating subtle, unexpected perturbation
- ✓ Adversarial training exposes the model to a blend of perturbed and clean data, acclimatizing it to varied input types.



[1] B. D. Son et al., "Adversarial Attacks and Defenses in 6G Network-Assisted IoT Systems" in *IEEE Internet of Things Journal* **Q1-IF =10.6**

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VI. Conclusion

QUICK SUMMARY

- Investigating adversarial attacks and defences for 6G network-assisted IoT systems.
- We propose three algorithms to attack complicated autoencoder.
- The experiment results show that the proposed algorithms are effective to deconstruct AE, while defense approaches also are discussed.
- The proposed algorithms must improve to be suitable for black-box scenarios.

MANY THANKS FOR YOUR ATTENTION!

[1] B. D. Son et al., "Adversarial Attacks and Defenses in 6G Network-Assisted IoT Systems" in *IEEE Internet of Things Journal* **Q1-IF = 10.6**, H-INDEX: 179

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[2] B. D. Son and authors., "Dynamic Offloading for Edge Computing-Assisted Metaverse Systems," in *IEEE Communications Letters* **Q1-IF = 4.1**, H-INDEX: 164

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[2] **B. D. Son** *and authors.*, "Dynamic Offloading for Edge Computing-Assisted Metaverse Systems," in *IEEE Communications Letters* **Q1-IF = 4.1**

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